The Taco Truck Decision

A small-business decision • Calculus 1

Diego runs a popular taco truck outside the engineering building. Business is booming — maybe *too* well. At lunch the line gets so long that students give up and leave, and Diego is wondering whether it's time to buy a **second truck**. A second truck is a big loan, so he wants to decide with numbers, not a hunch. He needs to know two things: **is one truck already too small for the lunch rush**, and **will demand grow enough to keep two trucks busy**? His records are below.

DIEGO'S RECORDS (everything you need)

The lunch rush	During the rush, customers arrive at the rate $\mathbf{C(t) = 120t - 30t^2}$ (people per hour), where t = hours after 11 a.m. (the rush runs 11–3). Diego can serve at most **100 people per hour**. Whenever arrivals run faster than that, a line builds and customers walk away.
Looking ahead	Diego serves **300 customers/day** today, and that's growing. The growth rate is $\mathbf{g(t) = 80 - 4t}$ (extra customers/day added each month), where t = months from now. A second truck only pays off if he'll be serving at least **600 customers/day within a year** (12 months).

Your two questions

Question 1. Is one truck already too small for the rush?

Find the **busiest moment** of the lunch rush and the peak arrival rate. (Hint: C(t) is an upside-down parabola — the peak is where the slope $C'(t)$ equals zero.) Compare that peak to Diego's limit of 100 people/hour. At the busiest moment, is he serving everyone or losing customers?

Question 2. Will demand justify a second truck?

The **extra** customers Diego gains over the next 12 months is the **integral of the growth rate**: $\int_0^{12} (80 - 4t)\, dt$. Add that to today's 300 to get his daily customers one year from now. Does it reach the 600 he needs? *Careful:* if you assume he just keeps adding 80/month forever, you'd get 300 + 80×12. Why does that overcount?

Write Diego a 2–3 sentence recommendation: should he buy the second truck, and what do the numbers say?

ANSWER KEY (instructor)

Q1. $C'(t) = 120 - 60t = 0 \rightarrow t = 2$, the busiest moment is **1 p.m.** Peak rate $C(2) = 240 - 120 =$ **120 people/hour**. Since 120 > 100, at the peak customers arrive faster than Diego can serve them — about 20/hour give up and leave. *One truck is already too small. (Skill: maximum via the derivative.)*

Q2. $\int_0^{12} (80 - 4t)\, dt = [80t - 2t^2]_0^{12} = 960 - 288 =$ **672 extra customers/day**. One year out: 300 + 672 = **972 customers/day**, well past the 600 threshold — demand justifies the truck. The flat-rate guess (300 + 80×12 = 1,260) overcounts because the growth rate $g(t) = 80 - 4t$ shrinks every month; a changing rate can't be treated as constant. *(Skill: turning a rate into a total with the integral / Net Change Theorem.)*

Verdict. Both signals say **expand**: Diego is already turning customers away at the 1 p.m. peak, and demand will climb to about 972/day within a year. He should buy the second truck.